FEDERICO COLOMBO

Computer Graphics Engineer • Technical Artist

Specialized in **real-time rendering**, low-level **engine architecture (C++)**, and **procedural pipelines**. M.Sc. in Cinema & Media Engineering at Politecnico di Torino (**Expected Graduation: September 2026**), with active R&D in bio-adaptive volumetric SDF raymarching and mobile XR optimization.



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WORK EXPERIENCE

Technical Artist & Producer (XR Master Thesis R&D)
03/2026 - Present

Lemons in the Room • Florence / Remote

- Architected a real-time bio-adaptive volumetric raymarching framework in HLSL for Meta Quest 3, rendering dynamic Signed Distance Fields (SDFs) driven by BLE biometric telemetry.
- Engineered GPU optimizations (Dynamic Bounding Volumes, adaptive ray stepping) locking **90 FPS on standalone XR hardware** (61.3% GPU time reduction, 0B GC alloc).
- Authored Morten Mikkelsen Surface Gradient shading on procedural volumes to eliminate triplanar UV seams, plus a sub-millisecond GPU Voxelizer for particle swarms.

Technical Artist, Gameplay Programmer & PM
11/2024 - 07/2026

LEVEL UP LAB (PoliTo Student Team) • Turin, Italy

- Authored custom shaders in HLSL (retro CRT curvature, water displacement) and programmed core C# gameplay across 2 productions (*Line Operator* & *Spicy Mercante*).
- Coordinated multidisciplinary pods of 6–7 student developers as Technical Producer and PM, enforcing Git LFS versioning and milestone sprints.

Software Developer & AI Consultant
06/2026 - Present

Intellecto IP • Turin, Italy / Remote

- Engineered a centralized personal banking web app in React, consolidating 4+ banking API feeds into a secure, reactive dashboard.
- Integrated Anthropic Claude LLM workflows into business processes, automating 15+ weekly operational analysis tasks via Notion API.

Teaching Assistant (GUI & Frontend)
10/2024 - 01/2025

Politecnico di Torino • Turin, Italy

- Tutored 60+ engineering students in UI/UX principles and frontend programming, achieving a 95% lab submission pass rate.
- Authored technical documentation and comprehensive instructional modules to improve laboratory comprehension.

Video Artist & Post-Production Specialist
09/2023 - 06/2024

RKH SRL • Turin, Italy

- Operated cinema camera systems (RED, Blackmagic 6K) across 12+ commercial productions, coordinating multi-point lighting setups.
- Managed DaVinci Resolve post-production workflows, accelerating corporate media turnaround timelines by 25%.

EDUCATION

M.Sc. in Cinema & Media Engineering
2023 - Sep 2026 (Expected)

Politecnico di Torino • **Expected: September 2026**

Focus: Real-Time Rendering, Game Engine Architecture, Computer Vision, Digital Signal Processing. Thesis: *PulseEngine — Volumetric Raymarching & Dynamic SDFs in XR.*

B.Sc. in Cinema & Media Engineering
2020 - 2023

Politecnico di Torino • Turin, Italy

Linear Algebra, Multivariate Calculus, Discrete Mathematics, Algorithms & Data Structures, C/C++ OOP.

CORE COMPETENCIES

Real-Time Rendering & Shaders

Compute shaders, Raymarching & SDFs, PBR equations, Deferred shading, SPOM, HLSL/GLSL.

Engine Architecture (C++)

Custom OpenGL engine development, memory layout, draw-call batching, multi-threading.

XR & Standalone Optimization

Meta Quest 3 (Snapdragon XR2), foveated raymarching, 90 FPS locked latency budgets.

DCC & AI Integration Pipelines

Python tooling, Claude LLM integration, Unreal PCG, automated asset workflows.

SOFTWARE & TOOLS

C++ (17/20, STL)	Advanced
HLSL / GLSL	Compute/Ray
Unity 6 (URP, C#)	Expert
Unreal Engine 5	PCG, Niagara
Blender & Python API	Add-ons
Python 3	Pipeline
RenderDoc & Nsight	Profiling
Git & Git LFS	CI/VCS

LANGUAGES

Italian	Native Speaker
English	Professional Working (C1)